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In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import random
```

```
In [3]: random.seed(42)
np.random.seed(42)
nrows=1000
data = {
    'SID': [i+1 for i in range(nrows)],
    'Name': [f'Student{i+1}' for i in range(nrows)],
    'Age': np.random.randint(15, 21, size=nrows),
    'Mathscore': np.random.randint(20, 100, size=nrows),
    'Sciencescore': np.random.randint(20, 100, size=nrows),
    'Englishscore': np.random.randint(20, 100, size=nrows),
    'Historyscore': np.random.randint(20, 100, size=nrows),
    'Geographyscore': np.random.randint(20, 99, size=nrows),
    'Attendance': np.random.randint(50, 100, size=nrows),
}
df = pd.DataFrame(data)
sub = ['Mathscore', 'Sciencescore', 'Englishscore', 'Historyscore', 'Geographyscore']
df['Percentage'] = df[sub].mean(axis=1).astype(int)
def grade(percentage):
    if percentage >= 90:
        return 'A'
    elif percentage >=80:
        return 'B'
    elif percentage >=70:
        return 'C'
    elif percentage >=60:
        return 'D'
    elif percentage >= 33 & percentage <60:
        return 'P'
    else:
        return 'F'
df['Grade'] = df['Percentage'].apply(grade)
df.to_csv('Student1.csv', index = False)
```

```
In [4]: df = pd.read_csv('Student1.csv')
```

```
In [5]: df.head(2)
```

```
Out[5]:
```

	SID	Name	Age	Mathscore	Sciencescore	Englishscore	Historyscore	Geographyscore	Attendance	Percentage	Grade
0	1	Student1	18	41	45	92	29	33	54	48	P
1	2	Student2	19	40	27	80	54	97	97	59	P

```
In [6]: for col in df.columns:
        if col not in ['SID', 'Age', 'Name', 'Grade', 'Attendance']:
            missing = random.sample(range(nrows), k=int(nrows*0.1))
            df.loc[missing, col] = np.nan
```

```
In [7]: df.head(2)
```

```
Out[7]:
```

	SID	Name	Age	Mathscore	Sciencescore	Englishscore	Historyscore	Geographyscore	Attendance	Percentage	Grade
0	1	Student1	18	41.0	45.0	NaN	29.0	33.0	54	48.0	P
1	2	Student2	19	40.0	27.0	80.0	54.0	97.0	97	59.0	P

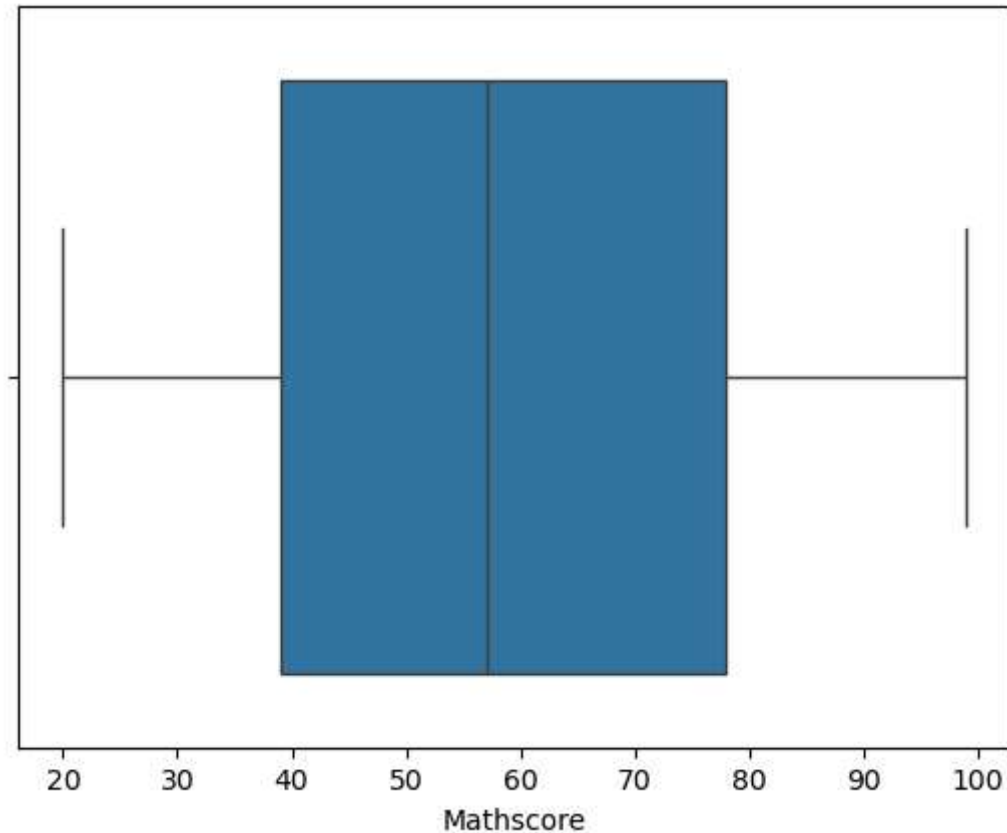
```
In [8]: df.isnull().sum()
```

```
Out[8]: SID          0
        Name          0
        Age           0
        Mathscore    100
        Sciencescore 100
        Englishscore 100
        Historyscore 100
        Geographyscore 100
        Attendance    0
        Percentage    100
        Grade         0
        dtype: int64
```

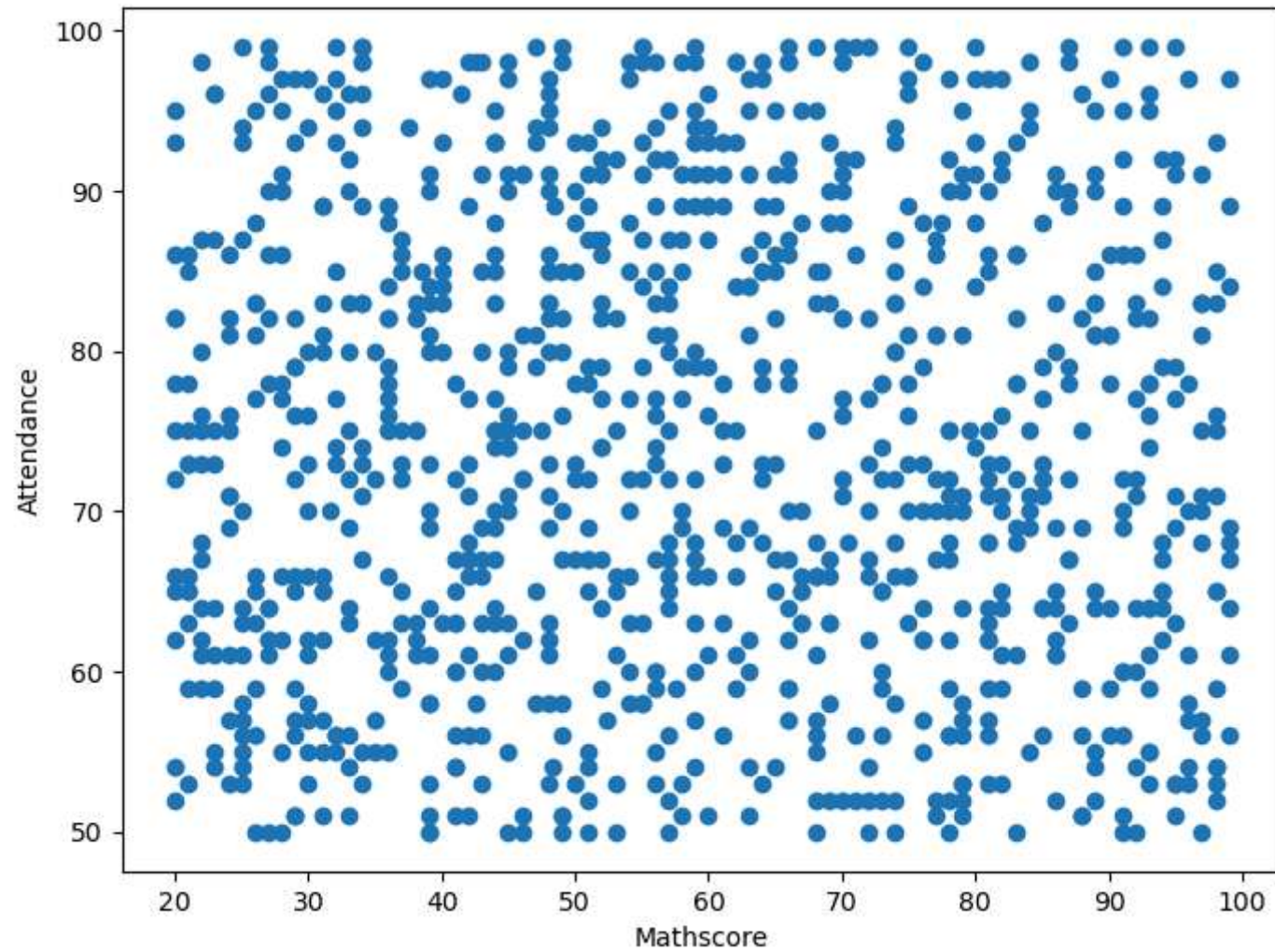
```
In [9]: df['Attendance'] = df['Attendance'].apply(lambda x: int(x[:-1]) if isinstance(x, str) and x != '' else x)
```

```
In [10]: sub = ['Mathscore', 'Sciencescore', 'Englishscore', 'Historyscore', 'Geographyscore']  
df[sub] = df[sub].apply(lambda x: x.fillna((df.loc[x.name, 'Percentage']*5 - x.sum())/x.isna().sum())  
                        if x.isna().any() else x, axis=1)
```

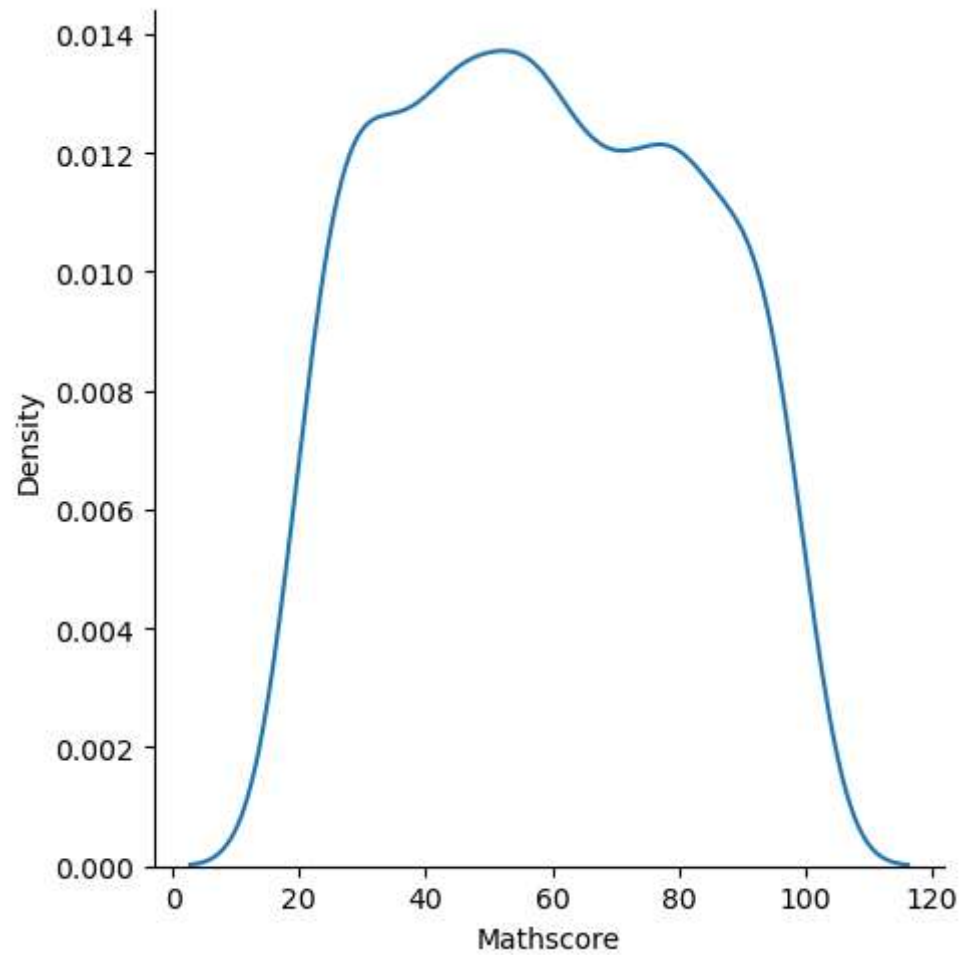
```
In [12]: sns.boxplot(x=df['Mathscore'])  
plt.show()
```



```
In [16]: fig, ax = plt.subplots(figsize=(8,6))  
ax.scatter(df['Mathscore'], df['Attendance'])  
ax.set_xlabel('Mathscore')  
ax.set_ylabel('Attendance')  
plt.show()
```



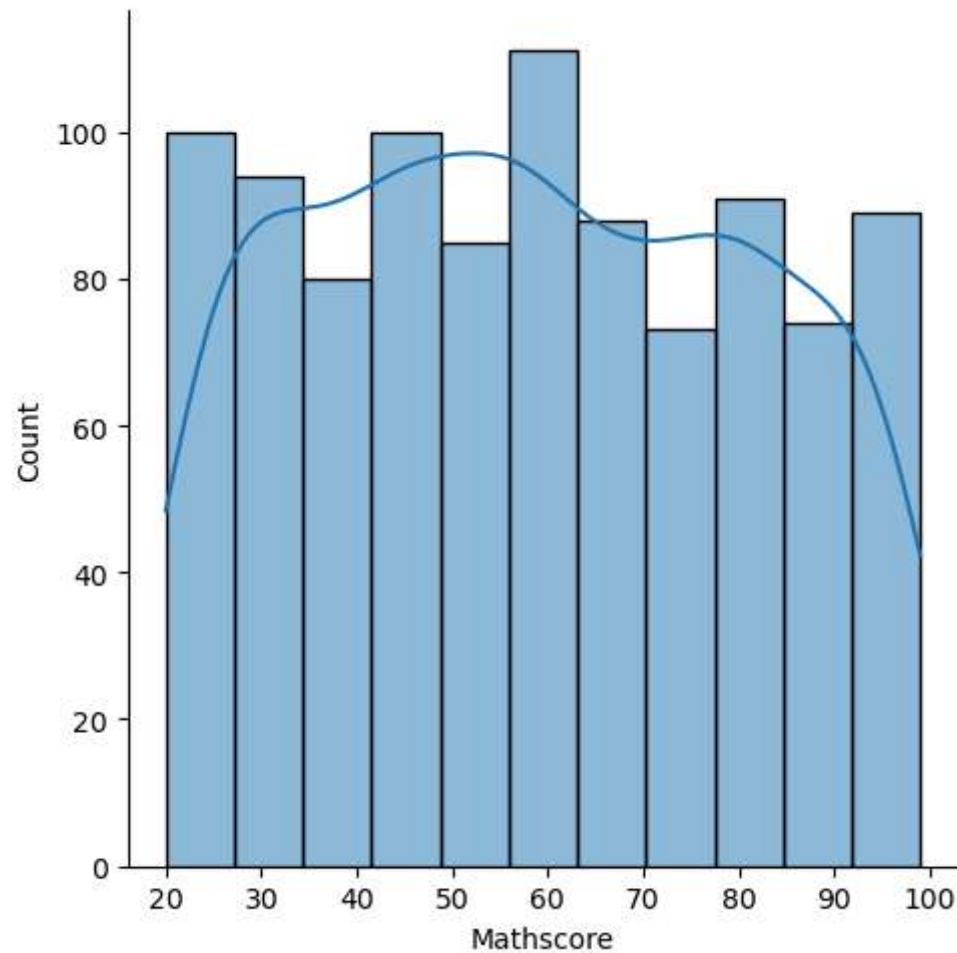
```
In [17]: sns.displot(x='Mathscore', data = df, kind='kde')  
plt.show()
```



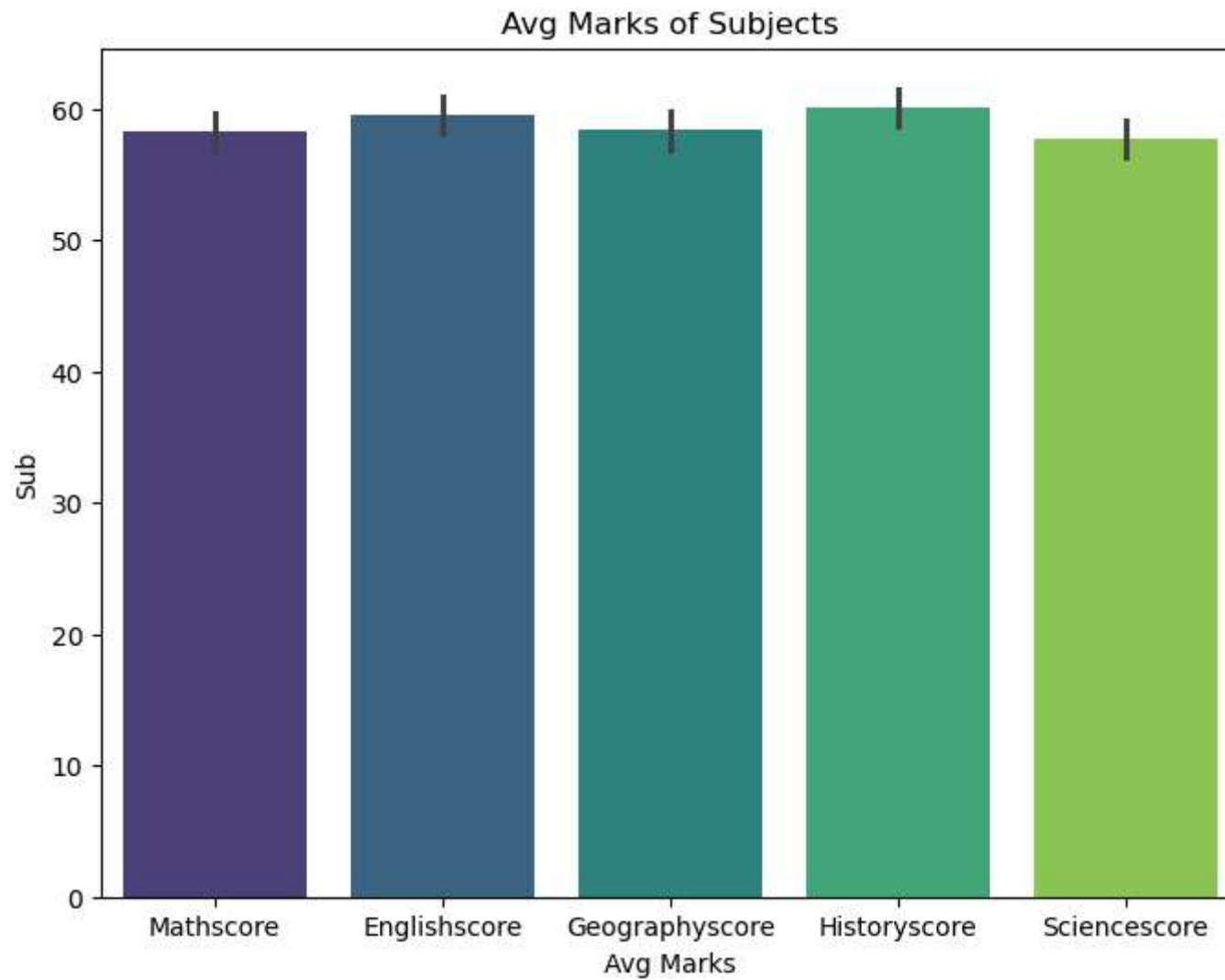
```
In [18]: df['Mathscore'].skew()
```

```
Out[18]: 0.06728203963605163
```

```
In [19]: sns.displot(x='Mathscore', data=df, kde=True)  
plt.show()
```



```
In [20]: sub = ['Mathscore', 'Englishscore', 'Geographyscore', 'Historyscore', 'Sciencescore']
dfmelt = df[sub].melt(var_name = 'Sub', value_name = 'Marks')
plt.figure(figsize=(8,6))
sns.barplot(x='Sub', y='Marks', hue='Sub', data=dfmelt, estimator=np.mean, palette='viridis')
plt.title('Avg Marks of Subjects')
plt.xlabel('Avg Marks')
plt.ylabel('Sub')
plt.show()
```



```
In [21]: def findoutliers(df, cols):  
    outliers = {}  
    for col in cols:  
        q1 = df[col].quantile(0.25)  
        q3 = df[col].quantile(0.75)  
        iqr = q3 - q1  
        lb = q1 - 1.5*iqr  
        ub = q3 + 1.5*iqr
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    outliers[col] = df[(df[col]<lb) | (df[col]>ub)]
    return outliers

```

```
In [29]: outlier_res = findoutliers(df, ['Mathscore', 'Sciencescore', 'Englishscore', 'Geographyscore', 'Historyscore', 'Atter
```

```
In [30]: for col, outlier in outlier_res.items():
    print(f'{col}: {len(outlier)} outliers')
```

```

Mathscore: 0 outliers
Sciencescore: 0 outliers
Englishscore: 0 outliers
Geographyscore: 0 outliers
Historyscore: 0 outliers
Attendance: 0 outliers

```

```
In [31]: def outlierdetection(df,cols,threshold=3):
    outliers = {}
    for col in cols:
        zscore = (df[col] - df[col].mean()) / df[col].std()
        outliers[col] = df[abs(zscore) > threshold]
    return outliers
```

```
In [32]: outlier_zscore = outlierdetection(df,['Mathscore', 'Sciencescore', 'Englishscore', 'Geographyscore', 'Historyscore',
```

```
In [33]: for col, out_df in outlier_zscore.items():
    print(f'{col}: {len(out_df)} outliers')
```

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Mathscore: 0 outliers
Sciencescore: 0 outliers
Englishscore: 0 outliers
Geographyscore: 0 outliers
Historyscore: 0 outliers
Attendance: 0 outliers

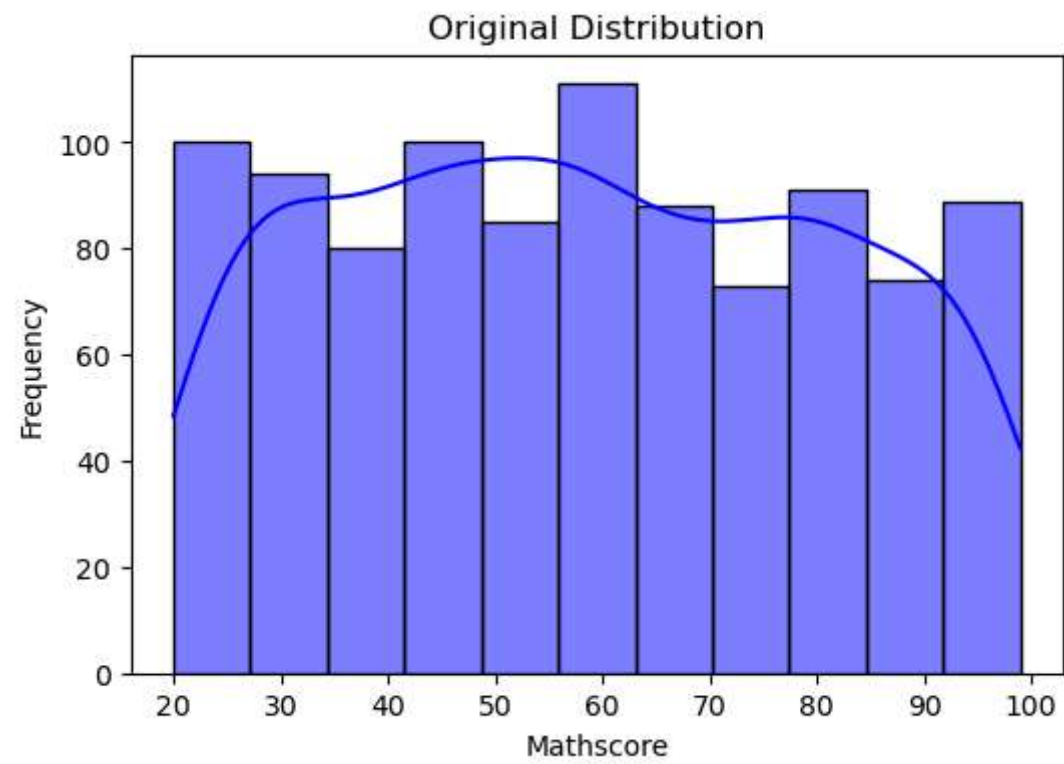
```

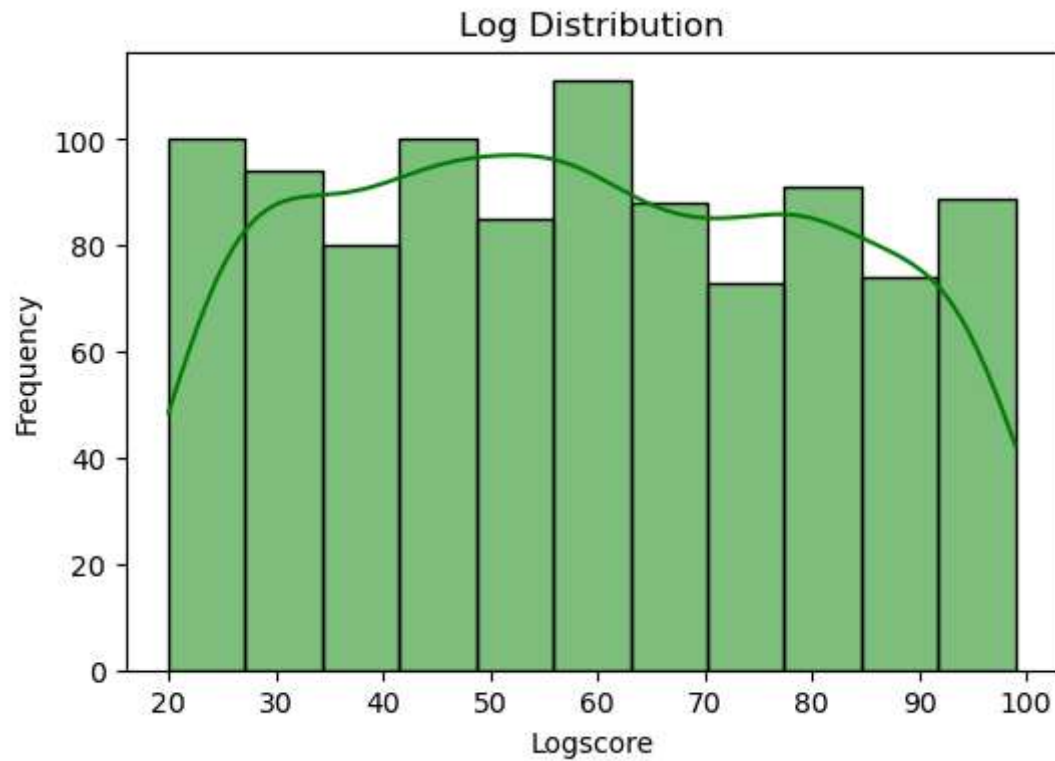
```
In [34]: df['Mathscore'] = df['Mathscore'].apply(lambda x:1 if x<=0 else x)
df['Logscore'] = np.log(df['Mathscore'])
df[['Mathscore', 'Logscore']].head(2)
```


Out[34]:

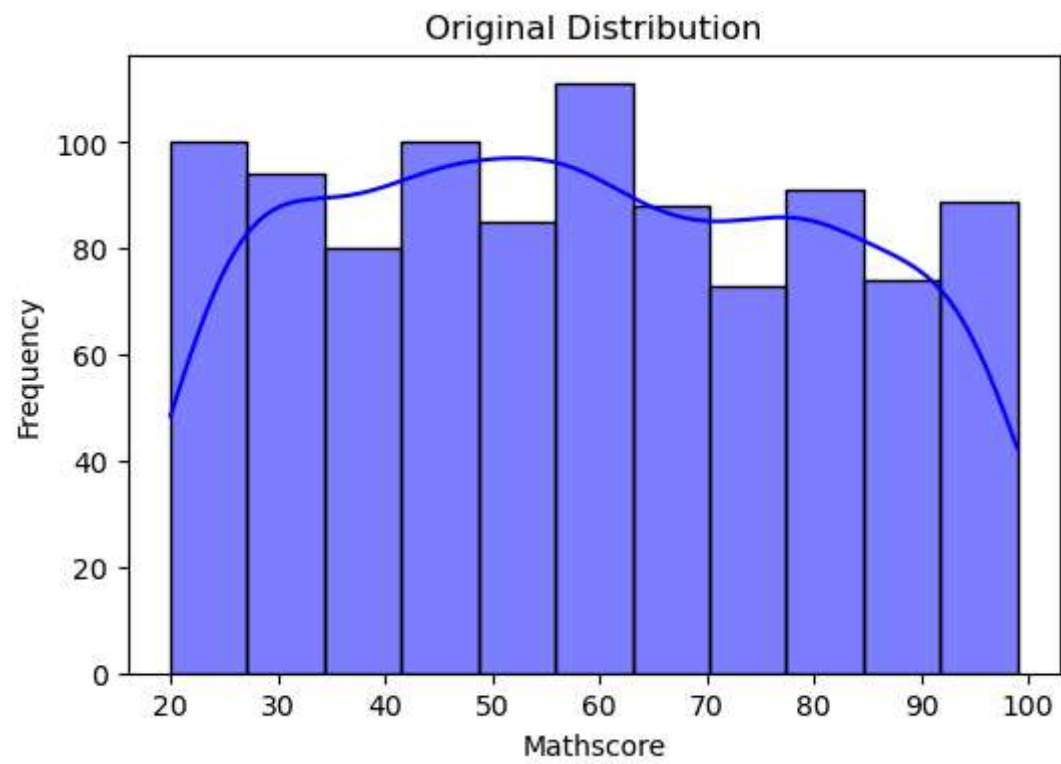
	Mathscore	Logscore
0	41.0	3.713572
1	40.0	3.688879

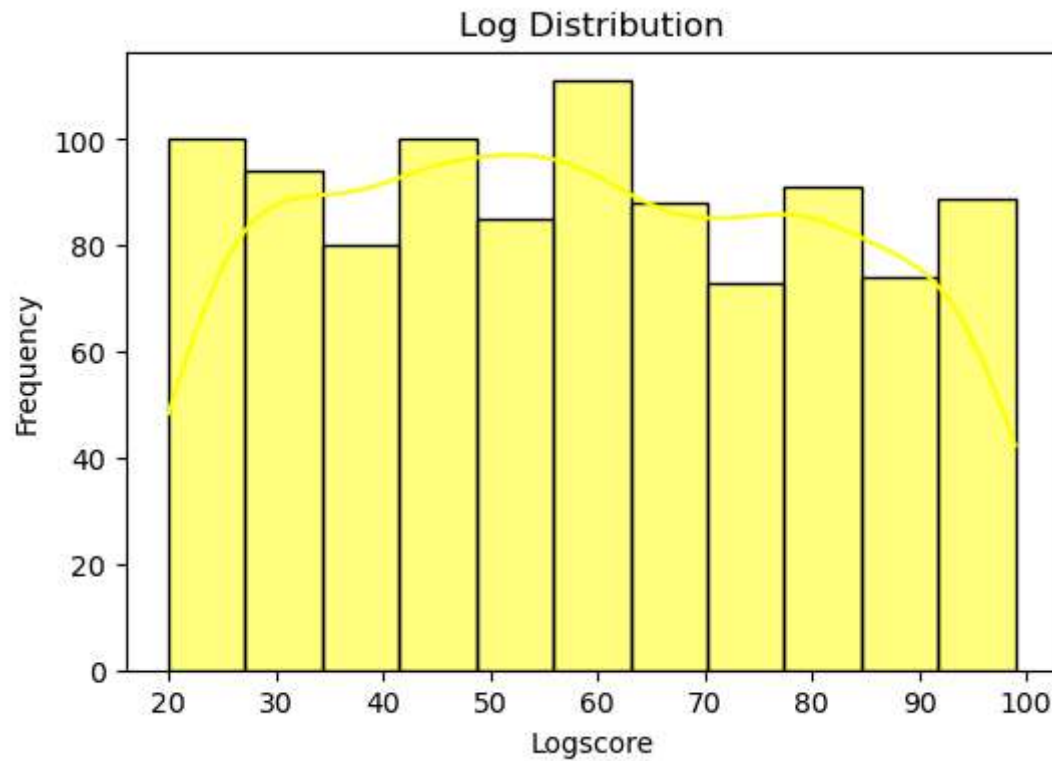
```
In [35]: plt.figure(figsize=(6,4))
sns.histplot(df['Mathscore'], kde=True, color='blue')
plt.title('Original Distribution')
plt.xlabel('Mathscore')
plt.ylabel('Frequency')
plt.show()
df['Logscore'] = np.log(df['Mathscore']+1)
plt.figure(figsize=(6,4))
sns.histplot(df['Mathscore'], kde=True, color='green')
plt.title('Log Distribution')
plt.xlabel('Logscore')
plt.ylabel('Frequency')
plt.show()
```



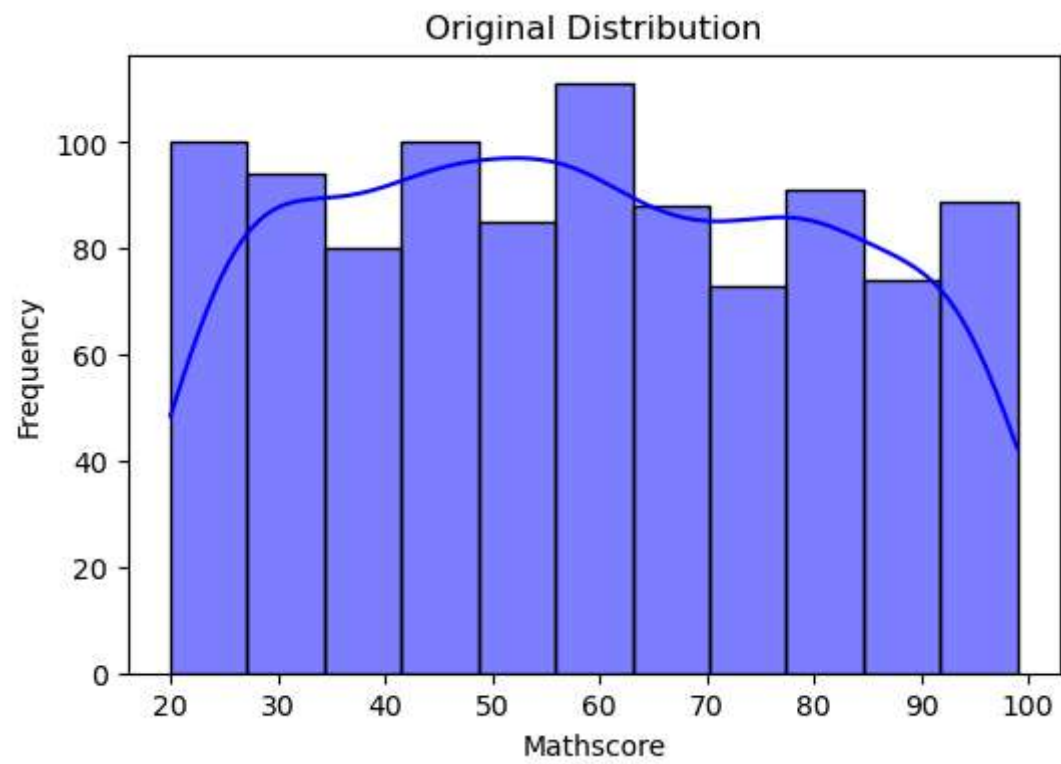


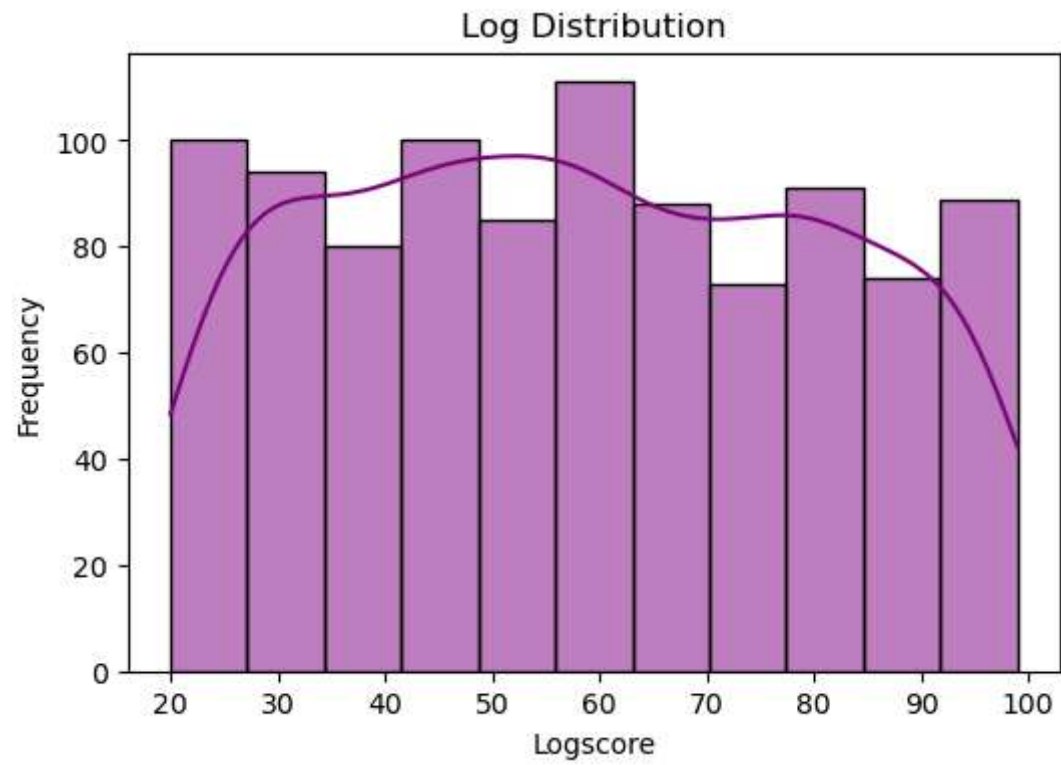
```
In [36]: plt.figure(figsize=(6,4))
sns.histplot(df['Mathscore'], kde=True, color='blue')
plt.title('Original Distribution')
plt.xlabel('Mathscore')
plt.ylabel('Frequency')
plt.show()
df['Logscore'] = df['Mathscore'] ** 0.5
plt.figure(figsize=(6,4))
sns.histplot(df['Mathscore'], kde=True, color='yellow')
plt.title('Log Distribution')
plt.xlabel('Logscore')
plt.ylabel('Frequency')
plt.show()
```





```
In [37]: plt.figure(figsize=(6,4))
sns.histplot(df['Mathscore'], kde=True, color='blue')
plt.title('Original Distribution')
plt.xlabel('Mathscore')
plt.ylabel('Frequency')
plt.show()
df['Logscore'] = df['Mathscore'] ** (1/3)
plt.figure(figsize=(6,4))
sns.histplot(df['Mathscore'], kde=True, color='purple')
plt.title('Log Distribution')
plt.xlabel('Logscore')
plt.ylabel('Frequency')
plt.show()
```





In []: